# Practice workbook · Apply the triplet code to bases, codons and amino acids

Converting DNA triplets to mRNA codons and tRNA anticodons, reading amino acids from a table, counting bases per amino acid, and counting possible codes as a power.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-APPLY-TRIPLET-CODE

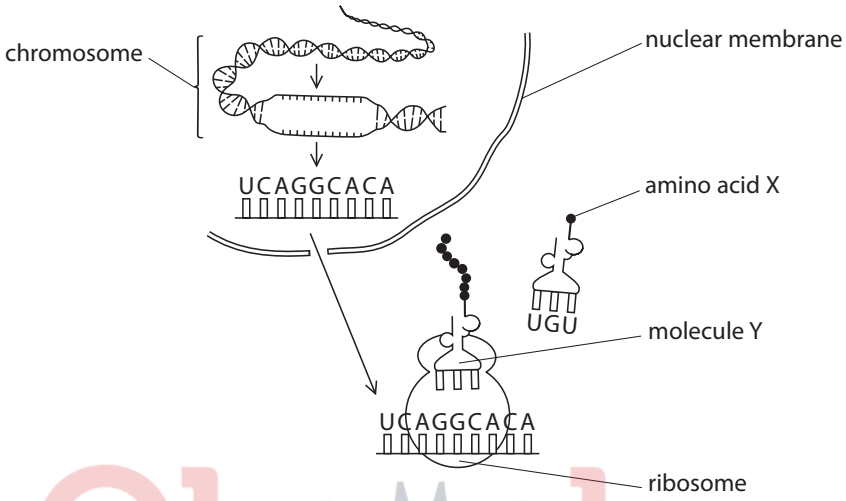
# 1. Question 6 · part c · 1 mark

June 2019 Paper 1R

Worked-example retry · Attempt part c only.

**6** Protein synthesis has a number of stages.

The diagram shows some of the stages.




The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

| Base triplet of DNA | Amino acid |
|---|---|
| AGT | serine |
| CCG | glycine |
| TGT | threonine |
| GTA | histidine |
| CAA | valine |
| TCC | argenine |
| ACA | cysteine |

(a) Describe the function of the nuclear membrane.

**(2)**

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# 1. Question 6 · part c · 1 mark

June 2019 Paper 1R

Worked-example retry · Attempt part c only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

**(1)**

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

**(1)**

(d) Describe the process occurring on the ribosome.

**(6)**

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## 1. Question 6 · part c · 1 mark · Working and answer

Working and answer

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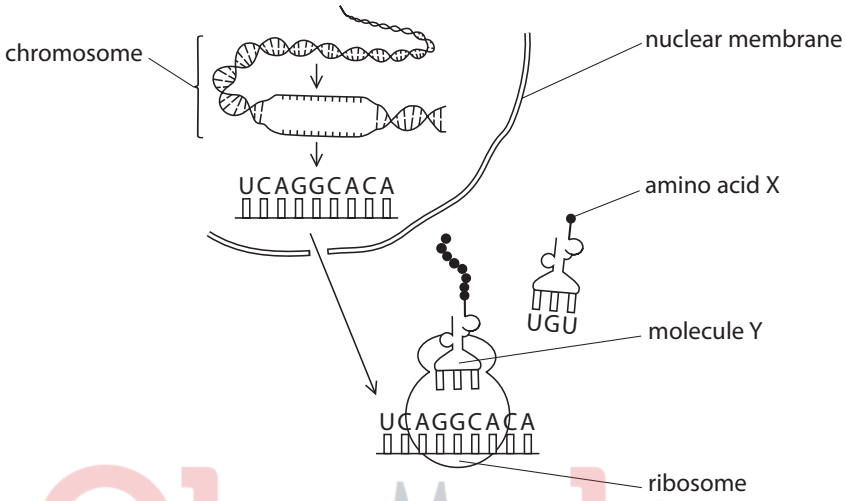
## 2. Question 6 · part b · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b only.

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June 2019 Paper 1R

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## 2. Question 6 · part b · 1 mark · Working and answer

Working and answer

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## 3. Question 8 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

**8** (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

**(2)**

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

**(2)**

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

**(2)**

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

**(2)**

number of nucleotides = .....

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## 3. Question 8 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

(v) Describe the structure of a nucleotide found in a DNA molecule.

**(3)**

(b) There are 20 different amino acids that can be joined to form a polypeptide chain.

The table gives some base sequences that code for five of these amino acids.

| Code | Name of amino acid |
|---|---|
| AAT or AAC | asparagine |
| TAT or TAC | tyrosine |
| TTT or TTC | phenylalanine |
| CGT or CGG | arginine |
| CAT or CAC | histidine |

Diagram 1 shows part of the gene that codes for a CFTR protein that functions normally.

Diagram 2 shows the same part of the gene with a mutation that causes cystic fibrosis.

AATATCATCTTTGGTGTTTCCTATGAT

**Diagram 1**

AATATCATCGGTGTTTCCTATGAT

**Diagram 2**

Describe the changes that occur in the gene and the protein that cause cystic fibrosis.

**(3)**

**(Total for Question 8 = 14 marks)**

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## 3. Question 8 · part a.iv · 2 marks · Working and answer

Working and answer

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# 4. Question 8 · part b · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

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## 4. Question 8 · part b · 3 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

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**Diagram 1**

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**Diagram 2**

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**(3)**

**(Total for Question 8 = 14 marks)**

## 4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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# 5. Question 1 · part c · 2 marks

June 2023 Paper 2R

Further practice · Attempt part c only.

**1** (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

**(3)**

| Cell structure | Function | | |
|---|---|---|---|
| | **makes proteins** | **contains DNA** | **allows substances in and out of cells** |
| nucleus | | | |
| ribosomes | | | |
| cell membrane | | | |

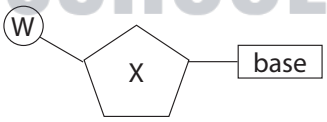
(ii) Give a structure found in bacterial cells.

**(1)**

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.




Name part W and part X.

**(2)**

W .....

X .....

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## 5. Question 1 · part c · 2 marks

June 2023 Paper 2R

Further practice · Attempt part c only.

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(ii) Describe how bases hold the two chains of nucleotides together.

**(3)**

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(c) The gene for the hormone insulin contains 153 bases.

Determine the number of amino acids in insulin.

**(2)**

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number of amino acids = .....

**(Total for Question 1 = 11 marks)**

## 5. Question 1 · part c · 2 marks · Working and answer

Working and answer

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## 6. Question 5 · part a · 2 marks

November 2023 Paper 1

Further practice · Attempt part a only.

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**5** DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

**(2)**

number of codes = .....

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

**(3)**

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(ii) State **one** factor that can cause mutation.

**(1)**

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

**(2)**

## 6. Question 5 · part a · 2 marks · Working and answer

Working and answer

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